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### Light emitting diode package, divisional display module, and display panel

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#### Abstract

A light emitting diode (LED) package includes a substrate, at least one micro LED chip, a black material layer, and a transparent material layer. The substrate has a width ranging from 100 micrometers to 1000 micrometers. The at least one micro LED chip is electrically mounted on a top surface of the substrate and has a width ranging from 1 micrometer to 100 micrometers. The black material layer covers the top surface of the substrate to expose the at least one micro LED chip. The transparent material layer covers the at least one micro LED chip and the black material layer.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 18/172,283, filed on Feb. 21, 2023, which is a continuation of U.S. application Ser. No. 17/308,070, filed on May 5, 2021, now U.S. Pat. No. 11,616,173, issued Mar. 28, 2023, which is a continuation of U.S. application Ser. No. 16/600,577 filed on Oct. 14, 2019, now U.S. Pat. No. 11,038,088, issued Jun. 15, 2021, which is related to copending U.S. application

Ser. No. 16/541,132 filed on Aug. 14, 2019, copending U.S. application Ser. No. 16/524,165 filed on Jul. 29, 2019 and copending U.S. application Ser. No. 16/524,202 filed on Jul. 29, 2019, all of which are incorporated by reference herein in their entireties and not admitted to be prior art with respect to the present disclosure by their mention in this cross-reference section.

## BACKGROUND

### Field of Invention

(1) The present disclosure relates to a light emitting diode package.

### Description of Related Art

(2) It is known to use LED displays each including an array of red, green, and blue LED elements as pixels. Such LED displays offer light with higher luminance than the luminance of backlight-type liquid-crystal displays and are used for large-scale digital signage and other applications.

(3) There is a demand for such LED displays to have high contrast ratios, but reflection of extraneous light may cause false lighting and reduce the contrast ratios in environments.

(4) Therefore, there is a need to provide a light-emitting device that offers higher luminance and a higher contrast ratio.

## SUMMARY

(5) One aspect of the present disclosure provides a light emitting diode package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate has a width ranging from 100 micrometers to 1000 micrometers. The plurality of micro LED chips are electrically mounted on a top surface of the substrate and has a width ranging from 1 micrometer to 100 micrometers. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. Each micro LED chip has a thickness substantially equal to that of the black material layer. The transparent material layer covers the plurality of micro LED chips and the black material layer. The micro LED chip includes a first semiconductor layer with a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and a supporting breakpoint on the light emitting surface. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

(6) Another aspect of the present disclosure provides a light emitting diode (LED) package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate having a width ranging from 100 micrometers to 1000 micrometers. The plurality of micro LED chips are electrically mounted on a top surface of the substrate and has a width ranging from 1 micrometer to 100 micrometers. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. Each micro LED chip has a thickness substantially equal to that of the black material layer. The transparent material layer covers the micro LED chips and the black material layer. The transparent material layer has a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

(7) Another aspect of the present disclosure provides a light emitting diode (LED) package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate having a width. The plurality of micro LED chips is electrically mounted on a top surface of the substrate. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED

chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. The plurality of micro LED chips has a thickness substantially equal to that of the black material layer. The transparent material layer covers the micro LED chips. The transparent material layer has a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4. The micro LED chip includes a first semiconductor layer with a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and a supporting breakpoint on the light emitting surface. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

- (1) The invention can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1 illustrates a cross-sectional view of a typical light emitting diode package.
- (3) FIGS. 2-5 illustrate cross-sectional views of light emitting diode packages in accordance with embodiments of the present invention.
- (4) FIG. 6 illustrates a top view of the light emitting diode package in accordance with another embodiment of the present invention.
- (5) FIG. 7 illustrates a cross-sectional view of a light emitting diode package in accordance with another embodiment of the present invention.
- (6) FIG. 8 illustrates a cross-sectional view of a micro LED chip in accordance with another embodiment of the present invention.
- (7) FIG. 9 illustrates a cross-sectional view of a micro LED chip in accordance with still another embodiment of the present invention.
- (8) FIGS. 10-11 illustrate steps of manufacturing a light emitting diode display panel in accordance with one embodiment of the present invention.
- (9) FIGS. 12-13 illustrate steps of manufacturing a light emitting diode display panel in accordance with another embodiment of the present invention.
- (10) FIG. 14 illustrates a cross-sectional view of a light emitting device according in accordance with another embodiment of the present invention.
- (11) FIG. 15 illustrates a cross-sectional view of a light emitting diode structure according in accordance with another embodiment of the present invention.
- (12) FIG. 16 is a photo of micro LED chip arrays of which each chip is equipped with a laser-lift-off rough pattern.
- (13) FIG. 17 is a photo of a top surface of a micro LED chip equipped with a laser-lift-off rough pattern.

### **DETAILED DESCRIPTION**

- (14) The present disclosure is described by the following specific embodiments. Those with ordinary skill in the arts can readily understand the other advantages and functions of the present invention after reading the disclosure of this specification. The present disclosure can also be implemented with different embodiments. Various details described in this specification can be modified based on different viewpoints and applications without departing from the scope of the present disclosure.
- (15) The following embodiments are disclosed with accompanying diagrams for detailed description. For illustration clarity, many details of practice are explained in the following

descriptions. However, it should be understood that these details of practice do not intend to limit the present invention. That is, these details of practice are not necessary in parts of embodiments of the present invention. Furthermore, for simplifying the drawings, some of the conventional structures and elements are shown with schematic illustrations.

(16) FIG. 1 illustrates a cross-sectional view of a typical light emitting diode package. A LED package **100a** includes a substrate **102a**, at least one mini LED chip **106a** and a transparent material layer **104a**. The mini LED chip **106a** conventionally includes a sapphire substrate SA. The transparent material layer **104a** covers the at least one mini LED chip **106a** and the substrate **102a**. In this embodiment, the substrate **102a** has a width W1 ranging from 100 micrometers to 1000 micrometers. The mini LED chip **106a** has a width ranging from 100 micrometers to 250 micrometers.

(17) In an embodiment A of a package width of 1000 micrometers (i.e., the substrate **102a** has a width of 1000 micrometers), a LED chip **106a** with a thickness of 150 micrometers and a width of 225 micrometers, and a transparent glue layer (e.g., **104a**) with a thickness equal to or greater than 250 micrometers, but no black material is covered over the substrate **102a**, the LED chip **106a** may emit to enable the package **100a** to achieve a 18% ratio of a side emission SE1 to a top emission TE1. In this embodiment, a ratio of the width (W1) of the substrate **102a** to a thickness of the transparent material layer **104a** is smaller than 4.

(18) As shown in FIG. 2, a LED package **100b** includes a substrate **102b**, at least one micro LED chip **106b** and a transparent material layer **104b**. In this embodiment, the substrate **102b** has a width W1 ranging from 100 micrometers to 1000 micrometers. The substrate **102b** is used to package a micro LED chip **106b** having a width W2 ranging from 1 micrometers to 100 micrometers and a thickness T2 smaller than 10 micrometers, instead of a mini LED chip, e.g., **106a**. The transparent material layer **104b** covers the at least one micro LED chip **106b** and the substrate **102b**.

(19) In certain embodiments, the substrate **102b** has a width W1 ranging from 100 to 200 micrometers, from 200 to 500 micrometers, from 500 to 750 micrometers, or from 750 to 1000 micrometers.

(20) In certain embodiments, the micro LED chip **106b** has a width W2 ranging from 1 micrometer to 100 micrometers, e.g., from 1 to 5 micrometers, from 5 to 10 micrometers, from 10 to 25 micrometers, or from 25 to 50 micrometers.

(21) In an embodiment B of a package width of 1000 micrometers (i.e., the substrate **102b** has a width of 1000 micrometers), a LED chip **106b** with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., **104b**) with a thickness T1 of 100 micrometers, but no black material is covered over the substrate **102b**, the LED chip **106b** may emit to enable the package **100b** to achieve a 5% ratio of a side emission SE2 to a top emission TE2.

(22) Comparing the embodiments A and B, the side emission in embodiment B is reduced because the micro LED chip **106b** is used to replace the mini LED chip **106a** in the package and the transparent glue layer is downsized in its thickness, thereby increasing an internal reflection within the transparent glue layer.

(23) As shown in FIG. 3, a LED package **100c** includes a substrate **102c**, at least one micro LED chip **106c** and a transparent material layer **104c**. In this embodiment, the substrate **102c** has a width W3 ranging from 100 micrometers to 1000 micrometers. The substrate **102c** is used to package a micro LED chip **106c** having a width W2 ranging from 1 micrometers to 100 micrometers and a thickness T2 smaller than 10 micrometers. The transparent material layer **104c** covers the at least one micro LED chip **106c** and the substrate **102c**.

(24) In an embodiment C of a package width of 400 micrometers (i.e., the substrate **102c** has a width of 400 micrometers), a LED chip **106c** with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., **104c**) with a thickness T1 of 100 micrometers, but no black material is covered over the substrate **102c**, the LED chip **106c** may emit to enable the

package **100c** to achieve a 10% ratio of a side emission SE3 to a top emission TE3.

(25) Comparing the embodiments B and C, the side emission in embodiment C is increased because the package or the substrate **102c** is downsized in its width to reduce an internal reflection within the transparent glue layer.

(26) As shown in FIG. 4, a LED package **100d** includes a substrate **102d**, at least one micro LED chip **106d** and a transparent material layer **104d**. In this embodiment, the substrate **102d** has a width W3 ranging from 100 micrometers to 1000 micrometers. The substrate **102d** is used to package a micro LED chip **106d** having a width ranging from 1 micrometer to 100 micrometers and a thickness T2 smaller than 10 micrometers. The transparent material layer **104d** covers the at least one micro LED chip **106d** and the substrate **102d**.

(27) In an embodiment D of a package width of 400 micrometers (i.e., the substrate **102d** has a width of 400 micrometers), a LED chip **106d** with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., **104d**) with a thickness T3 of 50 micrometers, but no black material is covered over the substrate **102d**, the LED chip **106d** may emit to enable the package **100d** to achieve a 4% ratio of a side emission SE4 to a top emission TE4.

(28) Comparing the embodiments C and D, the side emission in embodiment D is reduced because the transparent glue layer **104d** is downsized in its thickness to further increase an internal reflection within the transparent glue layer.

(29) As shown in FIG. 4, the at least one micro LED chip **106e** has a thickness substantially equal to that of the black material layer **108**, but not being limited thereto.

(30) In other embodiments, the transparent glue layer **104d** may have a top text surface **104d'** to further increase the top emission TE4 for the LED package **100d**.

(31) As shown in FIG. 5, a LED package **100e** includes a substrate **102e**, at least one micro LED chip **106e**, a black material layer **108** and a transparent material layer **104e**. In this embodiment, the substrate **102e** has a width W3 ranging from 100 micrometers to 1000 micrometers. The substrate **102e** is used to package a micro LED chip **106e** having a width ranging from 1 micrometer to 100 micrometers and a thickness smaller than 10 micrometers. The black material layer **108** is configured to cover a top surface of the substrate **102e** and expose a light-emitting surface of the at least one micro LED chip **106a**. The black material layer **108** preferably has a thickness less than 10 micrometers.

(32) In an embodiment E of a package width of 400 micrometers (i.e., the substrate **102e** has a width of 400 micrometers), a LED chip **106e** with a thickness of 10 micrometers and a width of 50 micrometers, a black material layer **108** with a thickness 3 micrometers and a reflectivity smaller than 10%, and a transparent glue layer (e.g., **104e**) with a thickness T3 of 50 micrometers, the LED chip **106e** may emit to enable the package **100e** to achieve a 0.4% ratio of a side emission SE5 to a top emission TE5. In this embodiment, the black material layer **108** may be a black photoresist having a reflectivity smaller than 10%, but not being limited thereto.

(33) Comparing the embodiments D and E, the side emission in embodiment E is further reduced because the black material layer **108** is added to reduce internal reflection within the transparent material layer **104e**.

(34) In embodiments B-E, the transparent material layer may have an optical transmittance greater than or equal to 90%, 92%, or 95%, and have a thickness smaller than 100 micrometers, but not being limited thereto.

(35) In embodiments B-E, a ratio of the width (W1, W3) of the substrate (**102a-102e**) to the thickness (T1, T3) of the transparent material layer (**104a-104e**) is equal to or greater than 4 to suppress the side emission from the LED package such that the ratio of the side emission to the top emission can be reduced.

(36) In embodiments B-E, a width of a LED chip may be referred as a longer edge of the LED chip or any edge of the LED chip in a square shape while a width of a substrate may be referred as a longer edge of the substrate or any edge of the substrate in a square shape, but not being limited



thereto.

(37) In embodiments B-E, the at least one LED chip may include multiple LED chips configured to emit different color lights, e.g., red, green, blue lights, but not being limited thereto. Further, by adding cyan or yellow to the red, green, and blue micro LED, it can broaden the color gamut.

(38) FIG. 6 illustrates a top view of a light emitting diode package in accordance with another embodiment of the present invention. The light emitting diode package includes a substrate **102** and at least one LED chip **106** electrically mounted on a top surface of the substrate **102**. The substrate **102** has a top surface with an area **A1** while the LED chip **106** has its light emitting surface with an area **A2**. A ratio of the area **A2** to the area **A1** is preferably equal to or less than 5% to enhance the contrast ratio for the LED package such that the ratio of the side emission to the top emission can be reduced. For example, the ratio of the area **A2** to the area **A1** is 1% in embodiment B; the ratio of the area **A2** to the area **A1** is 5% in embodiments C, D, and E. In embodiments B-E, at least one LED chip **106** may be multiple LED chips, e.g., red, green, and blue LED chips, and the area **A2** may be referred as a total sum area of the light emitting surfaces of all the LED chips. In case the top surface of the substrate **102** is fully covered by the black material layer **108** except the light emitting surface of the LED chip **106**, the ratio of the side emission to the top emission may be further reduced, the contrast ratio may be further enhanced, and extraneous light may not be reflected to cause false lighting.

(39) The ratio of the side emission to the top emission, as discussed in previous embodiments, is reduced to improve crosstalk issues between pixels on the LED display panel.

(40) FIG. 7 illustrates a cross-sectional view of a light emitting diode package **100f** in accordance with another embodiment of the present invention. The LED package **100f** includes a substrate **102f**, at least one micro LED chip **106f** and a transparent material layer **104f**. In this embodiment, the transparent material layer **104f** is a transparent dielectric layer instead of a transparent glue layer, e.g., a transparent resin layer. The transparent dielectric layer may be formed by a chemical vapor deposition process or an atomic layer deposition process, but not being limited thereto. The transparent dielectric layer may be  $\text{SiO}_2$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{TiO}_2$ ,  $\text{Ta}_2\text{O}_5$ ,  $\text{HfO}_2$ ,  $\text{ZrO}_2$ ,  $\text{Y}_2\text{O}_3$ ,  $\text{MgF}_2$  or  $\text{Si}_3\text{N}_4$ , but not being limited thereto. The transparent dielectric layer, formed by a chemical vapor deposition process or an atomic layer deposition process, may have a thickness ranging from 0.1 micrometer to 10 micrometers, e.g., from 0.1 to 2 micrometers, from 2 to 4 micrometers, from 4 to 6 micrometers, or from 6 to 10 micrometers such that it can be conformal over upper surfaces of the micro LED chip **106f** and the substrate **102f**. In other embodiments, a black material layer, e.g., **108** in FIG. 5, may be added to cover a top surface of the substrate **102f** and expose a light-emitting surface of the at least one micro LED chip **106f**, and be covered by the transparent material layer **104f**.

(41) FIG. 8 illustrates a cross-sectional view of a micro LED chip in accordance with another embodiment of the present invention. The micro LED chip includes a semiconductor stack **120'** and a supporting breakpoint BP. The semiconductor stack **120'** includes a first semiconductor layer **121'**, a light emitting layer **122'**, and a second semiconductor layer **123'**. The first semiconductor layer **121'** has a light emitting surface **S1** exposed outside and the light emitting surface **S1** has a rough texture. The light emitting surface **S1** is formed by applying a laser lift off process to a sapphire substrate or patterned sapphire substrate to obtain a laser-lift-off rough pattern. Please see photos in FIGS. 16-17 for a laser-lift-off rough pattern, i.e., uniform round or hexagonal concaves, on a patterned sapphire substrate. Therefore, the micro LED chip is not equipped with the sapphire substrate or patterned sapphire substrate, but equipped with the laser-lift-off rough pattern to enhance light extraction. Specifically, the LED chip may emit a light passed through the light emitting surface **S1**. The supporting breakpoint BP is on the light emitting surface **S1**. In some embodiments, the first semiconductor layer **121'** includes doped semiconductor layers **121b'**, **121c'** and an undoped semiconductor layer **121a'**, and the light emitting surface **S1** is on the undoped semiconductor layer **121a'**. The doped semiconductor layers **121b'**, **121c'** are between the light

emitting layer **122'** and the undoped semiconductor layer **121a'**. In some embodiments, the LED chip further includes a first conductive pad **191** and a second conductive pad **192**. The first conductive pad **191** is electrically connected to the first semiconductor layer **121'**, and the second conductive pad **192** is electrically connected to the second semiconductor layer **123'**. In some embodiments, the LED chip further includes a conductive contact layer **124'** between the second conductive pad **192** and the second semiconductor layer **123'**. A sacrificial layer **140** is formed on the light emitting surface **S1** of the semiconductor stack **120'**. The sacrificial layer **140** is etched to form a supporter **142** protruded from a base portion thereof. A carrier substrate **160** is formed over the sacrificial layer **140**. In some embodiments, an adhesive layer **170** may be formed between the sacrificial layer **140** and the carrier substrate **160**. The carrier substrate **160** can be adhered to the sacrificial layer **140** by the adhesive layer **170** to enhance an adhesion therebetween. A sidewall leakage reduction layer **182** is formed over a sidewall of the semiconductor stack **120'** by an atomic layer deposition process. Thus, the sidewall leakage reduction layer **182** reduces sidewall leakage effect and increases external quantum efficiency (EQE). In this embodiment, the sidewall leakage reduction layer **182** may be formed from Al.sub.2O.sub.3 or SiO.sub.2, but not being limited thereto.

(42) A bottom mirror layer **181** is formed over an outer surface of the sidewall leakage reduction layer **182** to enhance light extraction via the light emitting surface **S1**. In this embodiment, the bottom mirror layer **181** may be a distributed Bragg reflector (DBR) formed from alternately layers of SiO.sub.2 and TiO.sub.2, but not being limited thereto.

(43) The semiconductor stack **120'** can be separated from the carrier substrate **160**. e.g., a sapphire substrate, when the carrier substrate **160** is removed by breaking the supporting breakpoint BP. A detailed process for manufacturing the LED chip can be cross-referenced to the specifications of U.S. application Ser. No. 16/524,202 filed on Jul. 29, 2019.

(44) FIG. 9 illustrates a cross-sectional view of a micro LED chip in accordance with still another embodiment of the present invention. The semiconductor structure in this embodiment is different from that of FIG. 8 in that the supporting breakpoint BP is on a surface of the semiconductor stack **120'** that is opposite to the light emitting surface **S1**. The light emitting surface **S1** is formed by applying a laser lift off process to a sapphire substrate or patterned sapphire substrate to obtain a laser-lift-off rough pattern. Therefore, the micro LED chip is not equipped with the sapphire substrate, but equipped with the laser-lift-off rough pattern to enhance light extraction.

(45) In other embodiments, the micro LED chips (**106b-106f**) may include the light emitting devices (**100**, **100a**, **100b**, **100c**) as described in the specifications of U.S. application Ser. No. 16/524,165 filed on Jul. 29, 2019. For example, FIG. 14 illustrates the light emitting device **100** as disclosed in FIG. 1 of U.S. application Ser. No. 16/524,165. The light emitting device **100** has a stacked structure **102** and a first insulating layer **104** covering at least side surfaces of the stacked structure **102**. The stacked structure **102** includes a p-type semiconductor layer **106**, an n-type semiconductor layer **110**, and a light emitting layer **108**. The n-type semiconductor layer **110** is on the p-type semiconductor layer **106**. The light emitting layer **108** is sandwiched between the p-type semiconductor layer **106** and the n-type semiconductor layer **110**. In some embodiments, the p-type semiconductor layer **106** is a p-type GaP layer, and the n-type semiconductor layer **110** is an n-type AlGaInP layer. The stacked structure **102** further includes an n-type electrode **114**, an n-type contact layer **112**, a p-type electrode **118**, an n-type contact pad **116**, and a p-type contact pad **120**. The n-type electrode **114** is on the n-type semiconductor layer **110**. The n-type contact layer **112** is sandwiched between the n-type semiconductor layer **110** and the n-type electrode **114**. The n-type contact pad **116** is on the n-type electrode **114**. The p-type electrode **118** is on the p-type semiconductor layer **106**. The p-type contact pad **120** is on the p-type electrode **118**. In some embodiments, the n-type contact layer **112** is an n-type GaAs layer. In some embodiments, the light emitting layer **108** is a multiple quantum well active layer. In an embodiment, the multiple quantum well active layer is formed of alternating layers of a well layer and a barrier layer. In some

embodiments, the light emitting layer **108** emits red light and the light emitting device **100** is a red light emitting device. The stacked structure **102** further includes a semiconductor reflector **122** between the light emitting layer **108** and the n-type contact layer **112**. In some embodiments, the n-type semiconductor layer **110** has a first portion **110A** and a second portion **110B** spaced apart from the first portion **110A** by the semiconductor reflector **122**. The second portion **110B** of the n-type semiconductor layer **110** is closer to the n-type contact layer **112** than the first portion **110A** of the n-type semiconductor layer **110**. A normal projection of the light emitting layer **108** onto the second portion **110B** of the n-type semiconductor layer **110** overlaps a normal projection of the semiconductor reflector **122** onto the second portion **110B** of the n-type semiconductor layer **110**. The semiconductor reflector **122** can reflect light which is generated in the light emitting layer **108**. In particular, the semiconductor reflector **122** can prevent the n-type contact layer **112** from absorbing light which is generated in the light emitting layer **108**, thereby enhancing the light emission efficiency of the light emitting device **100**. In other words, the semiconductor reflector **122** can redirect the light from passing downwards (i.e., in a direction toward the n-type contact layer **112**) to passing upwards (in a direction toward the p-type semiconductor layer **106**). In some embodiments, the semiconductor reflector **122** is a distributed Bragg reflector (DBR). The semiconductor reflector **122** includes multiple periods. Each period includes at least a first layer **124** and at least a second layer **126**. A refractive index of the first layer **124** is different from a refractive index of the second layer **126**. The first layer **124** and the second layer **126** of the semiconductor reflector **122** include aluminum in some embodiments. The refractive index of the first layer **124** and the refractive index of the second layer **126** depend on the atomic percentage of the aluminum therein. For example, the first layer **124** of the semiconductor reflector **122** includes  $\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}$ , in which  $0 < x < 1$ . The second layer **126** of the semiconductor reflector **122** includes  $\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}$ , in which  $0 < y < 1$  and  $y$  is different from  $x$ . That is to say, an atomic percentage of the aluminum in the first layer **124** is substantially different from an atomic percentage of the aluminum in the second layer **126**. In some embodiments, the first insulating layer **104** covers a bottom surface of the stacked structure **102** and exposes a bottom part of the n-type contact pad **116** and a bottom part of the p-type contact pad **120**. The first insulating layer **104** has a refractive index less than a refractive index of the p-type semiconductor layer **106** such that light extraction efficiency can be improved by the first insulating layer. The p-type semiconductor layer **106** has a top surface **106a** facing away from the semiconductor reflector **122**. The top surface **106a** of the p-type semiconductor layer **106** has irregularities. In other words, the top surface **106a** of the p-type semiconductor layer **106** is a rough surface, thereby improving the light extraction efficiency of the light emitting device **100** by reducing loss due to total internal reflection (TIR) between air and the top surface **106a** of the p-type semiconductor layer **106**.

(46) In other embodiments, the micro LED chips (**106b-106f**) may include the light emitting diode structures (**10**, **20**, **30**) as described in the specifications of U.S. application Ser. No. 16/541,132 filed on Aug. 14, 2019. For example, FIG. **15** illustrates a light emitting diode structure as disclosed in FIG. 50 of U.S. application Ser. No. 16/541,132. Process steps for manufacturing the light emitting diode structure can be referred to FIGS. 39 to 49 of U.S. application Ser. No. 16/541,132.

(47) FIGS. **10-11** illustrate steps of manufacturing a light emitting diode display panel using the light emitting diode chip in FIG. **8**. As shown in FIG. **9**, after micro LED chips, e.g., blue LED chips **107B**, green LED chips **107G**, and red LED chips **107R**, are manufactured on their respective substrates (**152**, **154**, **156**), a temporary substrate **150** is used to serially support blue, green, red LED chips (**107B**, **107G**, **107R**) which are first mass transferred from the respective substrates (**152**, **154**, **156**) using stamps **153**. After the first mass transfer, the blue, green, red LED chips (**107B**, **107G**, **107R**) have their electrodes exposed, a pre-test before molding may be performed to screen out mal-functional LED chips. Next, the temporary substrate **150** attached with all blue, green, red LED chips (**107B**, **107G**, **107R**) in proper positions is then turned upside down to perform second mass transfer and mount all blue, green, red LED chips (**107B**, **107G**, **107R**) on a

circuit board **158** for further molding with a transparent material layer **162** to be cut along dashed lines and tested. Only functional LED packages **100'** are transferred and attached on a blue tape **200**, and mal-functional LED packages **100'** are screened out. It is noted that each LED package **100'** includes a cut and divided circuit board **158**. Each LED package **100** can be the LED packages **(100b-100f)** as discussed in previous embodiments. As shown in FIG. **11**, functional LED packages **100** are then mounted on a division circuit board **301** and covered by an encapsulation layer **302** to form a divisional display module **300** of a complete LED display panel **400**. In another embodiment, functional LED packages **100'** may be mounted on a division circuit board **301** without an encapsulation layer **302**. When a mini LED package **100** encapsulating at least one micro LED chip has a package size ranging from 100 micrometers to 1000 micrometers, mal-functional packages can easily be tested and screened out before mounting on a division circuit board. The divisional display module **300** with full functional LED packages **100'** can be achieved. In another embodiment, the division circuit board **301** may include a system controller to control LED packages **100'**, e.g., using a system controller to control a pixel controller included in each LED package. The system controller may be mounted on a top surface **301a** of the division circuit board **301** or on a bottom surface **301b** of the division circuit board **301**.

(48) FIGS. **12-13** illustrate steps of manufacturing a light emitting diode display panel using the light emitting diode chip in FIG. **9**. The steps in FIGS. **12-13** are different from steps in FIGS. **10-11** in that the micro LED chips, e.g., blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'** are mass transferred from the respective substrates **(152, 154, 156)** directly to the circuit board **158** via a single mass transfer (e.g., using stamps **153**) instead of two times mass transfers. Therefore, a pre-test before molding may not be performed on the LED chips, but the testing after molding are still available to screen out mal-functional LED packages before mounting on a division circuit board **301**. In this embodiment, each LED package **100** include a set of blue LED chip **107B'**, green LED chip **107G'**, and red LED chip **107R'**. In other embodiments, each LED package may include a pixel controller to control multiple sets of blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'**. For example, as indicated by M4, 4 sets of blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'** as well as a pixel controller may be packaged together to perform a predetermined function, but not being limited thereto. For example, 6, 9 or more sets of R, G, B LED chips may also be packaged together.

(49) In sum, the mini LED package encapsulating at least one micro LED chip as disclosed herein are configured to offer higher luminance, a higher contrast ratio and reduce cross talks. In addition, the mini LED package encapsulating at least one micro LED chip facilitates easily testing the micro LED chips and screening out mal-functional ones before molding into a final display panel.

(50) The present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein. It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

## Claims

1. A divisional display module, comprising: a division circuit board; a plurality of light emitting diode (LED) packages mounted on the division circuit board; and an encapsulation layer covering the LED packages and the division circuit board, wherein each LED package comprises: a substrate having a width; a plurality of micro LED chips electrically mounted on a top surface of the substrate, wherein each micro LED chip has a thickness smaller than 10 micrometers; a black material layer covering the top surface of the substrate and exposing the plurality of micro LED

- chips, wherein each micro LED chip has the thickness substantially equal to that of the black material layer; and a transparent material layer covering the plurality of micro LED chips, and the transparent material layer having a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4.
2. The divisional display module of claim 1, wherein the black material layer comprises a thickness of 3 micrometers and a reflectivity smaller than 10%.
  3. The divisional display module of claim 1, wherein the transparent material layer has a top texture surface.
  4. The divisional display module of claim 1, wherein each micro LED chip comprises: a first semiconductor layer having a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and a supporting breakpoint on the light emitting surface.
  5. The divisional display module of claim 1, wherein the plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip.
  6. The divisional display module of claim 1, wherein the width of the substrate ranges from 100 micrometers to 1000 micrometers.
  7. The divisional display module of claim 1, wherein the black material layer is in contact with a sidewall of each micro LED chip.
  8. The divisional display module of claim 1, wherein each micro LED chip comprises a light emitting surface having a first area, the top surface of the substrate has a second area, a ratio of the first area to the second area is equal to or smaller than 5%.
  9. The divisional display module of claim 1, wherein the thickness of the transparent material layer is smaller than 100 micrometers.
  10. The divisional display module of claim 1, wherein the transparent material layer is a transparent dielectric layer or a transparent resin layer.
  11. A light emitting diode (LED) display panel, comprising: a plurality of divisional display modules, wherein each divisional display module comprises: a division circuit board; a plurality of LED packages mounted on the division circuit board; and an encapsulation layer covering the LED packages and the division circuit board, wherein each LED package comprises: a substrate having a width; a plurality of micro LED chips electrically mounted on a top surface of the substrate, wherein each micro LED chip has a thickness smaller than 10 micrometers; a black material layer covering the top surface of the substrate and exposing the plurality of micro LED chips, wherein each micro LED chip has the thickness substantially equal to that of the black material layer; and a transparent material layer covering the plurality of micro LED chips, and the transparent material layer having a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4.
  12. The LED display panel of claim 11, wherein the black material layer comprises a black photoresist having a reflectivity smaller than 10%.
  13. The LED display panel of claim 11, wherein the transparent material layer is a transparent dielectric layer or a transparent resin layer.
  14. The LED display panel of claim 11, wherein each micro LED chip comprises: a first semiconductor layer having a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and a supporting breakpoint on the light emitting surface.
  15. The LED display panel of claim 11, wherein each LED package further comprises a pixel controller configured to control the plurality of micro LED chips.

16. The LED display panel of claim 15, wherein the division circuit board further comprises a system controller to control the pixel controller included in each LED package.
  17. The LED display panel of claim 11, wherein the transparent material layer has a thickness smaller than 100 micrometers.
  18. The LED display panel of claim 11, wherein the width of the substrate ranges from 400 micrometers to 1000 micrometers.
  19. The LED display panel of claim 13, wherein the transparent dielectric layer comprises  $\text{SiO}_2$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{TiO}_2$ ,  $\text{Ta}_2\text{O}_5$ ,  $\text{HfO}_2$ ,  $\text{ZrO}_2$ ,  $\text{Y}_2\text{O}_3$ ,  $\text{MgF}_2$  or  $\text{Si}_3\text{N}_4$ .
  20. The LED display panel of claim 13, wherein the transparent dielectric layer is formed by a chemical vapor deposition process or an atomic layer deposition process.
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